

RAMCO INSTITUTE OF TECHNOLOGY
Department of Computer Science and Engineering
Academic Year: 2019-2020(Odd Semester)

INNOVATIVE TEACHING PRACTICES

Degree, Semester& Branch: III Semester B.E. ECE

Course Code & Title: EC8393 Fundamentals of Data Structures in C

Name of the Faculty member: Mr. C.A.Yogaraja

Date & Time: 14.08.2019 & 10:30 A.M

Topic: Functions, Pointers and Dynamic Memory Allocation

Lecture No.: 31

Practice's followed: Code Debugging

"Debugging is twice as hard as writing the code. Therefore, if you write the code as cleverly as possible, you are, by definition, not smart enough to debug it."

To inculcate the analysing ability in students, Code Debugging makes students to analyse, think and come out with more number of solutions to run the given code without bugs. Students were provided a code with bugs, they have to

“Identify the defective / incorrect part in the program”

“Write the exact / correct code at the defective places”

“Verify that the program is out of bug.”

Group Activity on Bug Identification:

In the current IT era, restructuring the existing application is a common need in the business. In which it requires in depth knowledge about the particular application. Students are exposed to the similar kind of environment through the code debugging activity.

In this activity,

- Students are asked to form a group maximum of two members
- Each group is provided with a half done problem (Code with error).
- The team's objective is to understand and analyse the given problem and correct the code to get the desired output.

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Team Members : J. Ninel Best Costa (953613106040)
Set No : 3
Date : 14.03.19

```
#include <stdio.h>
#include <stdlib.h>
void swap(); void swap(*int, *int)
int main() {
    int a = 100;
    int b = 200;
    printf("Before swap, value of a : %d\n", a);
    printf("Before swap, value of b : %d\n", b);
    swap(&a, &b);
    printf("After swap, value of a : %d\n", a);
    printf("After swap, value of b : %d\n", b);
    return 0;
}

void swap(int *x, int *y)
{
    int t;
    t = *x;
    *x = *y;
    *y = t;
    return;
}
```

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Team Members : N. Shambhi Reddy (953613106044), K. Savath. (953613106087)
Set No : 2
Date :

```
#include <stdio.h>
struct Distance
{
    int feet, inch;
};
float inch;
int main()
{
    struct Distance d1, d2, sum;
    printf("Enter information for 1st distance\n");
    printf("Enter feet: ");
    scanf("%d", &d1.feet);
    printf("Enter inch: ");
    scanf("%f", &d1.inch);
    printf("\nEnter information for 2nd distance\n");
    printf("Enter feet: ");
    scanf("%d", &d2.feet);
    printf("Enter inch: ");
    scanf("%f", &d2.inch);
    sum.feet = d1.feet + d2.feet;
    sum.inch = d1.inch + d2.inch;
    /* If inch is greater than 12, changing it to feet. */
    if(sum.inch > 12.0)
    {
        sum.inch = sum.inch - 12.0;
        ++sum.feet;
    }
    printf("\nSum of distances = %d\''-%.1f''", sum.feet, sum.inch);
    return 0;
}
```

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Team Members : M. Sohin Kannan , 18.
Set No : 1
Date : 14/8/19

#include <math.h>

int main()
{

```
printf("Enter number of elements in the Array: ");
scanf("%d", &num);
ptr = (int*) malloc(num * sizeof(int));
if(ptr == )
{
printf("Error! Unable to allocate memory.");
}
printf("Enter elements of array: ");
for(i = 0; i < num; ++i)
{
scanf("%d", );
}
for(i = 0; i < num; ++i)
{
sum += ;
}
printf("Sum of Elements in the given array is : %d", sum);
free(ptr);
return 0;
}
```

#include <stdio.h>
int num, ~~arr~~ sum=0;
int *ptr;

if (ptr == NULL)

scanf("%d", ~~arr~~ ^{ptr+i});

sum += *ptr + i;

free(~~arr~~ ^{*ptr});

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Team Members : A. Suryakala (953618106065).
Set No : 5
Date :

```
#include <stdio.h>
#include <stdlib.h> #include <math.h>
#define PI 3.1415
#define circleArea(r) (PI*r*r)
```

int main()
{

```
int radius;
float area;
printf("Enter the radius: ");
area = circleArea(radius scanf("%d", &radius););
printf("Area = %.2f", area);
return 0;
```

OUTCOME:

Finally, students have able to discuss with the team members and found the errors and they have corrected the code to bug free.

Prepared By

Approved By